

K HARSHAVARDHAN

CYBER SECURITY ANALYST

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PROFESSIONAL SUMMARY

Cyber Security Analyst with hands-on experience in authorized web application security assessments, ERP and endpoint security, vulnerability assessment, Linux, embedded/IoT security, and automotive cybersecurity. Skilled in Python, Nmap, Wireshark, Burp Suite, SIEM, OWASP Top 10, IDS/IPS, memory forensics, ESP32, MQTT/TLS, and HMAC-based authentication. Published cybersecurity research and inventor on two Indian patent applications.

PROFESSIONAL EXPERIENCE

Cyber Security Analyst — SSEV SOFTSOLS | Apr 2026 – Present

- Performed authorized web application security assessments, supported ERP security enhancement, and implemented website access restrictions for endpoint security.
- Built a custom Linux operating system using Linux From Scratch (LFS), working with system components, packages, services, and Linux environments.
- Contributed to SAVATAR automotive cybersecurity using ESP32 for touch-based vehicle door/window control and embedded security requirements.
- Designed and tested an ESP32 smart door-lock controller using Flask → MQTT over TLS → ESP32 communication, with HMAC-SHA256 authentication and timestamp/nonce replay protection.
- Contributed to holographic-fan communication analysis and NVIDIA Jetson embedded applications; prepared technical documentation for cybersecurity and embedded-system solutions.

TECHNICAL SKILLS

Cybersecurity: Web Application Security, Vulnerability Assessment, OWASP Top 10, SIEM, IDS/IPS, Log Analysis, Network Security, IoT/Embedded Security

Tools: Burp Suite, Nmap, Wireshark, Remnux

Programming: Python, C, SQL, Java, HTML

Embedded & Systems: ESP32, MQTT, TLS, HMAC-SHA256, Linux, Ubuntu, Linux From Scratch (LFS), NVIDIA Jetson

PROJECTS

Automated Vulnerability Scanning Tool — Developed a web scanning tool to identify potential vulnerabilities, errors, and security risks and generate reports for remediation.

Memory Forensics — Analyzed volatile memory and memory dumps to investigate malicious activity, malware indicators, and unauthorized access.

Moving Target Defense for DDoS Mitigation — Designed a multi-server architecture using dynamic proxy shuffling, secure authentication, ChaCha20/AES encryption, attack detection, and filtering.

AI-Based Phishing Website Detection — Developed a Flask-based phishing URL detection platform using TF-IDF vectorization and machine-learning models.

RESEARCH & INTELLECTUAL PROPERTY

- IEEE Conference Paper — Moving Target Defense Framework for Resilient DDoS Mitigation using Dynamic Proxy Shuffling and ChaCha20 Encryption.
- AMLDA 2025 Conference Paper — AI-Based Phishing Website Detection using Machine Learning.
- Indian Patent Application No. 202541117014 — Intelligent Automated Waste Segregation System with IoT-Enabled Data Monitoring and Classification Mechanism.
- Indian Patent Application No. 202641017466 — ChaCha20 Encryption for Mitigating DDoS Attacks in Defense Network Architectures.

EDUCATION

B.Tech, Computer Science & Engineering — Cyber Security | VNR VJIET, Hyderabad | 2022–2026 | CGPA: 7.52/10

Higher Secondary: 91.6% | Secondary: 10/10

CERTIFICATIONS & LEADERSHIP

Certifications: Google Professional Cyber Security Course; CEH (EC-Council); Microsoft Cyber Security Industrial Training; EC-Council Cloud Computing; Cisco Introduction to Cybersecurity; ArIES IIT Roorkee/Coincent Cyber Security & Ethical Hacking.

Leadership: President — VJ Garuda Vigilance (Cybersecurity Club), VNR VJIET; led workshops, awareness sessions, competitions, and mentoring.